

The Astronaut Who Might Actually Get Us to Mars

The improbable journey of Franklin Chang Díaz, who immigrated to the U.S. at eighteen, became an astronaut, tied the record for most spaceflights, and now might hold the key to deep space travel.



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Franklin Chang Díaz at the Ad Astra offices on Thursday, December 21, 2017.

Photographs by Jeff Wilson

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We like to tell stories—often in blockbuster films and children’s books—that dramatize a direct path from the initial spark of an idea to its realization. Decades of hard work and failed attempts are condensed into a brief montage that concludes with triumph. Thomas Edison thought electricity could illuminate the world, so he developed the lightbulb. John Atanasoff was frustrated by the limitations of late-twenties calculators, so he devised his own and soon followed it up with the first electronic digital computer. President Kennedy said we’d go to the moon, so we poured money and manpower into a rocket, and Neil Armstrong took one giant leap for mankind.

Except, we know, the actual narrative is more complicated. Take that last example. When Kennedy gave his speech in May 1961, the genius rocket engineer Wernher von Braun and his team, which eventually found a home at the Marshall Space Flight Center, in Alabama, had been investigating, for several years, how to get a man on the moon. The problem: the two main approaches weren’t working. The first method, a direct ascent from the Earth to the moon, would require a gigantic vehicle—bigger than the *Saturn V*—hardly the type of thing that could launch or land nimbly. In the second design, NASA would send several components of a spacecraft into Earth’s orbit, where they would have to be assembled by astronauts before launching to the moon.

But not everyone was locked in to the favored approaches. Over at the Langley Research Center, in Virginia, a skeptical engineer named John Houbolt had latched on to a decades-old concept called lunar orbit rendezvous, which proposed something entirely different: a crew could travel into lunar orbit, detach a lightweight, “spider”-looking module, and fly that down to the moon; once the crew had explored the lunar surface, they would guide the module back up to the primary ship, dock the two sections together, and fly home. This, Houbolt came to believe, was the solution the engineers were looking for.

Many of his colleagues thought he was nuts. The method’s unconventionality and riskiness made them nervous, and attempts to promote LOR were

repeatedly squelched. At one meeting, in December 1960, the space-engineering pioneer Max Faget stood up in an audience of Houbolt's supervisors and peers and dismissed him, saying, "He doesn't know what he's talking about." And so, in November 1961, Houbolt went out on a limb. Ignoring NASA's chain of command, he wrote a nine-page letter dramatically introducing himself as a "voice in the wilderness" to NASA's number two official, Robert Seamans Jr. The message was notable for its audacity and irritated tone as much as for its content. "The thoughts expressed here may not be stated in as diplomatic a fashion as they might be, or as I would normally try to do, but this is by choice," he wrote. "Do we want to go to the moon or not?"

What followed was nothing short of a paradigm shift for the entire Apollo spaceflight program. Within two weeks, intrigued NASA officials began passing Houbolt's memo around, and in June 1962, von Braun and others announced that the agency was scrapping its plans. NASA would pursue Houbolt's proposal instead. "Why was there so much resistance to it?" Houbolt later asked. "The only thing I can come up with is the syndrome of NIH: not invented here."

Right now, in research centers across America, we are trying to get to Mars, and scientists and engineers are encountering all the hurdles you might expect for such a monumental task. One of the most worrying is this: on a chemical rocket—the kind we've always used for space travel—it would take a crew nine months to reach Mars. Upon arrival, they'd have to wait at least three months for the planets to realign, and the return journey would take another nine months. In this scenario, the crew would be exposed to cancer-causing galactic rays for almost two years. Obviously this is something we'd like to avoid.

So really, getting to Mars is only part of the problem. If we go, we've got to get there and back fast. Finding that speed has eluded engineers, because chemical rockets require fuel, and fuel adds weight that drags the ship down. A solution might necessitate ideas outside the mainstream. It might come from somewhere unexpected.

In a cavernous warehouse behind a strip mall about three miles from the Johnson Space Center, just south of Houston, a Costa Rica-born scientist and former NASA astronaut named Franklin Chang Díaz has been tinkering with an unorthodox idea for the past thirteen years. The lab for his eleven-person company, Ad Astra, is roughly the size of a Barnes & Noble. But in the center of the room, rather than rows of romance novels, there are three stacks of gadgets resembling hi-fi stereo components. A few long tables and office chairs are scattered about. Along the back wall is the main event: a forty-ton stainless-steel cylindrical chamber that sits like a thirteen-foot-tall beer keg tipped on its side. If Franklin is right, this is the engine that will take us to Mars.

One afternoon last June, Franklin, a wrinkleless 67-year-old with a boyish mop of dark hair, wandered out of the office lunchroom with a cup of coffee to check on his team. Ad Astra was conducting phase two of a three-phase test on the variable specific impulse magnetoplasma rocket (VASIMR) engine. To fulfill a \$9 million NASA contract, the company must produce one hundred hours of continuous power at 100 kilowatts. “This is a very tiny part of a long journey,” Franklin said. For this phase, Ad Astra was required to accumulate only one hundred nonconsecutive hours of engine time, so they stopped every few minutes to cool the engine and monitor the effects of the Houston humidity, which could damage the system.

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Nearby, a handful of casually dressed Ad Astra scientists scanned a table of computer monitors and chased down problems they hadn't predicted, like players at Whac-A-Mole. When I crawled up a ladder to peer into the porthole of the engine's vacuum chamber, a purple cone of light burst forth from an eighteen-inch-wide hole in the wall separating the vacuum chamber from the engine. “So that's the plasma firing right now,” Franklin explained.

If all goes according to plan, that's the stuff that will blast out the back of a ship, cartoonlike, and zap us through space.

Plasma, a multimillion-degree, electrically conducting gas, is the most common visible state of matter in the universe, though we rarely see it on Earth. (We encounter the other three fundamental states—liquids, solids, and gases—daily.) We know plasma best from our sun, which is composed almost entirely of this fourth state of matter. Engineers have been using plasma in small engines since the sixties, but no one has magnified its power with as much success as Franklin.

Listening to him talk about plasma is a little like hearing someone describe a spray of fairy dust in dry, scientific terms. He will concede that we *can* get to Mars with a chemical rocket. And we'd need its power to escape Earth's orbit; the VASIMR can't do that on its own. But once in space, he'll stress, the benefits of the VASIMR are significant. First, it is reusable, and therefore more economical than a one-off chemical booster. Second, a chemical rocket is basically a controlled explosion—flying one is like riding a cannonball through space; a plasma rocket can shift gears, and if something goes wrong, the crew can course-correct or turn the ship around. Finally, well, it's crazy fast. Four engines consuming two and a half megawatts of electricity each—about the wattage needed to power a Boeing 737 passenger jet—can increase the rocket's speed continuously in zero gravity. On a trip from Earth to Mars, it could reach 111,000 miles per hour, delivering a crew in a brisk sixty to ninety days and significantly reducing the risks to the astronauts' health.

A plasma test at Ad Astra.

When I first heard about the VASIMR test, I imagined something akin to the fifties-era propulsion research on Wallops Island, off the coast of Virginia, where rocket testing would knock antique plates off the walls of area houses and incite stampedes amongst the island's wild ponies. Yet as I watched the plasma—roughly as hot as the sun's corona—I could hear only the gentle pulsing of the cryogenic pumps working to maintain the chamber's vacuum. The rest of the machine was eerily silent. In space, it won't make any sound at all.

Phase two was ultimately a success, but Franklin admitted that the final hurdle, scheduled for this summer, will be nerve-racking. "You have an engine that is exhausting plasma the temperature of the sun," he said. None of which, by the way, is cheap. "When we have one hundred hours and *one second*, we will turn it off."

It would be a huge accomplishment, but Franklin and his team also know that its historical significance will depend on what happens next: minor

things like whether the funding will continue and whether a few key people are willing to embrace unconventional, potentially frightening ideas. If we are serious about Mars, Franklin is the voice in the wilderness, asking, with a pleasant smile that masks his impatience, *Do we want to get to Mars or not?*

For decades, Franklin has good-naturedly tolerated what he calls “the giggle factor.” *Going to Mars? With a plasma engine?* He doesn’t take the puzzled looks personally. He’s used to it. The suggestion that Franklin is impractical has dogged him for most of his life. After all, how many Costa Rican children graduate from high school, knowing little English, and ask their parents to send them to the United States so they can become astronauts?

It’s a story that most in his circle have heard by now. In 1968, on a cold and damp August morning in Escazú, eighteen-year-old Franklin rose early to catch a flight. Not all of his family could fit into their three-person Peugeot pickup truck, but his mom wasn’t keen on goodbyes anyway. Franklin’s sister held his baby brother on her lap, and his father, the son of a Chinese immigrant, gripped the wheel. In Franklin’s pocket, he carried \$50 and a one-way ticket to Hartford, Connecticut. He’d arranged to move in with his aunt, uncle, and seven cousins in a three-bedroom apartment. He’d managed to memorize just a few phrases, like “Hello, my name is Franklin” and “Where is the school?” When his relatives picked him up at the airport, he swung his bag into their car and told them he was moving to the U.S. to become an astronaut.

His family encouraged him, of course, and yet who could be surprised when he struggled those first few months? Though he’d already gotten his diploma in Costa Rica, he decided that a senior year at a U.S. high school would help him get into an American college, and rather than suffer through the recommended English-language-learner’s courses, he requested entry into Hartford High’s senior class—complete immersion into a regular classroom. Sitting in his English class those initial weeks gave him plenty of time to reevaluate his ambition. He was attempting to read and respond to Dickens and Shakespeare when he could barely ask how to catch a bus, and by the end of the first quarter, he’d failed all his classes.

Only a few months in, though, his grades started to improve, eventually putting him near the top of his class, and before graduation, his homeroom teacher asked him to meet some local scouts from the University of Connecticut. As he describes in his autobiography, *Dream's Journey*, he arrived at the interview in his usual coat and tie, carrying a load of books on rocketry and orbital mechanics. (Most days he smoked a pipe, a habit that caused some kids to mistake him for a teacher.) He must have seemed the nerdiest of nerds, a stock character out of the movies. When the curious scouts asked Franklin about his books, he wasn't sure if he should tell the truth. "I want to be an astronaut and rocket scientist, and that is why I came to this country," he finally confessed.

The scouts had found their ideal candidate. Franklin entered UConn that fall, and though he was still struggling with English, he managed to absorb even the most advanced scientific concepts. He plunged headlong into experimental research in atomic physics, and he became interested in plasma. His research "was not only fascinating," he wrote in his memoir, "it was just plain cool." After UConn, he worked on his doctorate at the Massachusetts Institute of Technology, where he began to study the limitations of the modern chemical engine: it required too much fuel, it was too heavy, and it could never reach deep space. Franklin also saw the need for something faster, and he figured that if he could meld that vision with actual experience in spaceflight, he could invent a solution. By 1977, married with a young child, he was finishing his thesis, and his career was ascendant. He submitted his application to the Astronaut Selection Program.

Months later, when he got a stock rejection letter from NASA, he was devastated. He reminded himself that he rarely succeeded on the first try, but he still sank into a depression. He started avoiding newspaper stories about NASA's growing shuttle program. To make matters worse, he and his wife split up. At one point, as he was barreling down the highway on his motorcycle, he fantasized about getting into a wreck.

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mumbled, trying to find his voice. “Yes.”

He found some solace in his work. He joined the Flight Dynamics and Control Division at MIT’s Draper Laboratory, where he began experimenting with ways to control plasma’s exhaust velocity and thrust. This was the stuff of *Star Trek* (naturally, his favorite show). What he realized was: when a ship needed more thrust—say, to escape a gravitational field or push itself out of orbit—a driver could spend more fuel; once the ship broke away from the gravitational field, the driver could shift gears to spend less fuel and continue accelerating. For Franklin, this notion was thrilling.

Two years passed, and he thought he had made peace with his NASA rejection. Then one day he spotted a tall, sandy-haired man in a blue flight suit walking down a hallway of Draper labs. Franklin recognized him right away as Jack Lousma, a space shuttle astronaut. After exchanging a quick nod with Lousma, Franklin returned to his desk wondering if he should give NASA another shot. He’d gained valuable research experience since his rejection, he’d reasoned, and he’d also become a U.S. citizen. In October 1979, when NASA announced its plans to accept a class of nineteen astronauts into its program, he filled out the application, sealed the envelope, and sent it off to Houston.

Seven months later, he received a call from Johnson Space Center’s director of flight operations, George Abbey. Initially thinking the call was a prank, Franklin, then thirty years old, paused and searched for some clue that one of his Draper colleagues was teasing him. Who knew that he wanted to fly in space? Was the caller stifling a laugh? Only when Abbey grew impatient did Franklin grasp what was happening. Would Franklin accept a position in the upcoming class of astronauts? “Yes,” he mumbled, trying to find his voice. “Yes.”

Franklin (far right) on a *Discovery* mission, in 1998.

Courtesy of NASA/Newsmakers

Franklin on an *Endeavor* space shuttle mission, in 2002.

Courtesy of Franklin Chang Díaz

When Franklin landed in Houston, the astronaut culture at the JSC still seemed torn from the pages of *The Right Stuff*. Published around the same time as his arrival, the book famously captured the charisma and machismo of early NASA test pilots, and plenty of the pilots in Franklin's class came from a military tradition. He met guys like David Hilmers, a Marine who'd go on to fly four space shuttle missions, and David Leestma, a Navy test pilot who'd fly three.

Although Franklin no longer wore the suit and tie (and he'd quit the pipe), he was a different breed. He was at the forefront of a sea change emphasizing research at the agency. It even prompted a new entry in the astronaut training syllabus: "scientific and technical proficiency training," meaning

incoming scientists would continue to pursue their areas of expertise. But Franklin was no slouch as a pilot either. He eventually flew seven space shuttle missions and tied the record for most spaceflights with Jerry L. Ross. He logged more than 1,600 hours in space, including three spacewalks that totaled 19 hours and 31 minutes.

Franklin was impressed with NASA's newfound emphasis on recruiting scientists into the astronaut program, but he was surprised by ways that the agency was slow to change. When it came to Mars travel, many of his colleagues adamantly stuck to concepts they'd relied on to get to the moon. "Going to Mars is a leap in orders of magnitude and complexity," Franklin says. "NASA had fallen asleep or was enamored with the beauty of the Apollo program, because it was a wonderful thing, but it was a chapter that had to be closed, and we needed to open a new chapter that was different. That transition was not happening."

In 1981, NASA officials gave Franklin the okay to start a small research program at the JSC. But at meetings, in conferences, even in small talk, he could always sense skepticism about his plans for a plasma-propelled engine. And when doubt entered the room, resources found the exit. The major U.S. propulsion centers certainly didn't share his interest in high-power plasma: the Jet Propulsion Laboratory, in California, and the Glenn Research Center, in Ohio, continued to fund research into low-ion engines and their sibling, Hall thrusters; the Marshall Space Flight Center, in Alabama, focused on chemical propulsion.

The hesitation wasn't entirely rooted in an unfamiliarity with plasma. It was also based on anxiety—and it wasn't unfounded. The engine he was proposing would require considerable power. Solar energy would be sufficient in cislunar space—the expanse between the Earth and the moon. But any flight beyond the moon (farther away from the sun's rays) would require a nuclear reactor. That specification tended to provoke groans. "‘Nuclear power’ was almost a swear word," Franklin says. "And still is. There's a lot of taboos about nuclear power, but I have pointed out for years that without nuclear space power, we are not going to go far in space. No way."

To allay fears, Franklin often mentioned the U.S.S. *Nautilus*, the first nuclear submarine, built in 1955. The nuclear navy had a stellar record, he'd tell skeptics, and the ship was so well-shielded that passengers were exposed to more radiation just walking down the street than they were inside the submarine. (Even now, some researchers are uneasy. Jason Crusan, the director of NASA's Advanced Exploration Systems division, allows that putting a nuclear reactor aboard a spaceship would not be all that different from installing one in a submarine, "except that you're strapping it to a rocket, flying it through the atmosphere, and getting it into space, which could make some people who don't understand the technology nervous.")

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Franklin says some colleagues capitalized on the fear and the uncertainty to undermine his work. The sabotage took many forms. Hastily assembled review panels critiqued his research and didn't allow him to provide a rebuttal. Once, after a board temporarily shut down his lab, other researchers tried to poach his machinery for parts. Colleagues complained that he was working on a "dangerous nuclear device" and that he didn't have enough oversight. But perhaps the most brazen example: in 2002, Franklin had just blasted off on the space shuttle and was on his way to the International Space Station when a review board locked his team out of his laboratory and temporarily canceled its contract.

Decades of fighting for funding had finally taken a toll, and his frustration had peaked. "You're just one more mouth to feed, and they don't want to see you there," Franklin says with a shrug. "That's the law of the land." But who else would pay for him to work on plasma besides the government?

As it turned out, plenty of people. By 2004, the private space industry had grown, and a crop of new billionaires—the kinds of entrepreneurs who'd

struck it rich because of their own unorthodox ideas—were eyeballing Franklin's work. One day Elon Musk visited him at the JSC's lab, the Advanced Space Propulsion Laboratory. Later, Franklin chatted with one of the backers of the Ansari X Prize, a \$10 million award for the first private company to successfully launch a reusable manned spacecraft twice within a two-week period. The more people he spoke with, the more seriously Franklin began to consider if he too should go private. After all, the ASPL was starved for funding anyway. The officials holding the purse strings at the time seemed to think that plasma was technology for the distant future. "And I kind of felt like, 'We have to do it now in order to have it for the future,'" Franklin says. "That attitude was, to me, too narrow. Too slow. I needed to move on, to move fast."

By year's end, he'd decided to shift his laboratory out from under NASA's guardianship and, with its blessing, create his own company. His old NASA boss, George Abbey, whom Franklin had brought on as a board member, coined the name: Ad Astra, Latin for "to the stars." With his second wife's approval, Franklin pulled his family's entire liquid savings out of the bank: \$50,000. Next, he found a 24,000-square-foot warehouse near the JSC and began rolling a convoy of equipment there, past a series of chain restaurants and auto parts shops. Then Franklin left the agency he'd dreamed of joining as a boy and bet on himself.

Franklin at the Ad Astra offices in December 2017.

It is there, in that nondescript warehouse, that the future of planetary flight may be changed forever. One afternoon I met Franklin in the Ad Astra conference room, which is decorated with artistic renderings of futuristic-looking spaceships. Many have giant solar panels bigger than the actual ships. There's the solar-electric LEO Space Cleaner, for example, which would remove space debris that poses a threat to satellites and ships, and the Ocelot, an Ad Astra concept that would boost space platforms like the ISS back to orbit when they lose altitude. These are the kinds of vehicles that, in the short term, Franklin hopes to power with the VASIMR.

In the next five years, he envisions launching a remote-controlled, solar-powered plasma engine that can operate at about 100 to 200 kilowatts, enough power to send a cargo ship full of supplies to the space station. "The concept is to do a demonstration flight, probably take it for a spin around the moon or something like that," he says.

Mars remains his ultimate ambition, but Franklin will quickly point out that Ad Astra is first and foremost a commercial endeavor. “Mars is not really good business right now,” he admits. And so he hopes to break into the \$300 billion satellite and global communications market, becoming part of the trucking industry in near-space. In addition to hauling supplies to the space station and cleaning up space junk, his device could also be the engine for ships that, say, redirect an asteroid aimed at a metropolitan area.

These days, Franklin should have nothing left to prove, and nobody doubts his guts. As one stubbornly hopeful VASIMR supporter put it, “I cut light-years of slack to anybody who puts his ass on a million pounds of explosives seven times.” Still, he is driven by his doubters, and going private, Franklin has found, was one of the best decisions he’s ever made. In 2005, when a Swiss billionaire invested many millions—more than twice the amount of money Franklin had seen in his project’s 25 years at NASA—Ad Astra began making the costly transition from physics to engineering. “Physics is hard, but it’s not expensive,” he explains. “Engineering is not hard, but it’s expensive. And spaceflight is even more expensive.”

He imagines the Earth will become a protected area for humans, like a giant national park.

Ad Astra is not the only company making this shift. Two others, Aerojet Rocketdyne (of California) and MSNW (of Washington), are competing in the plasma arena, hoping to prove to NASA that their technology is best prepared for the next phase. Franklin doesn’t discount his competition, but he is confident that a company will want his invention eventually. And though he doesn’t specifically name SpaceX, Blue Origin, Orbital ATK, Boeing, or Lockheed Martin, those companies would clearly be interested in the results of his labor. Former NASA administrator Mike Griffin agrees. “I think that in the end, [Franklin’s engine] will carry the day,” he told me.

At the same time, Franklin objects to the perception, all too common, that traveling to Mars is irrational or somehow extravagant—that the costs far exceed the benefits—because he is anxious about our not-too-distant future. Many astronauts and space engineers gaze into the night sky, identifying with explorers like Lewis and Clark. Franklin stares into the heavens plenty, but just as often, he looks around in the midst of a Houston traffic jam and questions whether we'll be able to sustain a global population of ten billion people in the mid-century. "I think we'll need to relax this a little bit by letting humanity move out," he says. He imagines the Earth will become a protected area for humans, like a giant national park. "And we will always be able to come back to the beauty of the planet and preserve us, but probably not all of us will live here."

This is why, while plenty of people still roll their eyes at the prospect of a plasma-powered rocket to Mars or are scared off by the need to make use of nuclear power, Franklin believes that all of it is essential, however difficult. "I've been doing this for nearly forty years now. So yeah, it's grueling," he says. "My wife tells me that she doesn't know how I keep going, but I'm an optimistic guy. I've always been. And I feel that we still have a big mountain to climb."

In 1969, when Apollo 11 finally landed on the moon, John Houbolt was no longer at NASA. Yet his contribution was not forgotten. On July 20, officials invited him to the control room to watch the Eagle detach and touch down. The moment the module landed, everyone in the control center leaped from their chairs and cheered. It was the culmination of years of work and sweat from so many, and it might never have happened if not for one persistent voice. Wernher von Braun turned around to face Houbolt and made an "okay" sign. "Thank you, John," he said.

If Franklin were to choose the ending of his own story—the kind found in blockbuster films and children's books—he wouldn't simply attach his engine to a ship. He would go one step further. He would enroll in a refresher astronaut-training course, reacquainting his body with the hours it has spent in weightlessness. Then he would suit up and board the ship himself. His vehicle would be crazy-looking, no doubt, perhaps with giant petals of solar panels that would bloom once he reached cislunar space. He would then gaze

out the window and marvel at Earth's beauty. And he would pilot *his engine*. The VASIMR. His life's work.

He used to dream about the possibilities, about what it would feel like to fly that ship. These days he's not so sure it will happen. He's getting older, after all. He knows how it sounds. It seems a bit, well, impractical. "But, I mean, look at John Glenn. He was in his seventies and he flew," he says. "Maybe by some fluke I'll get another chance."